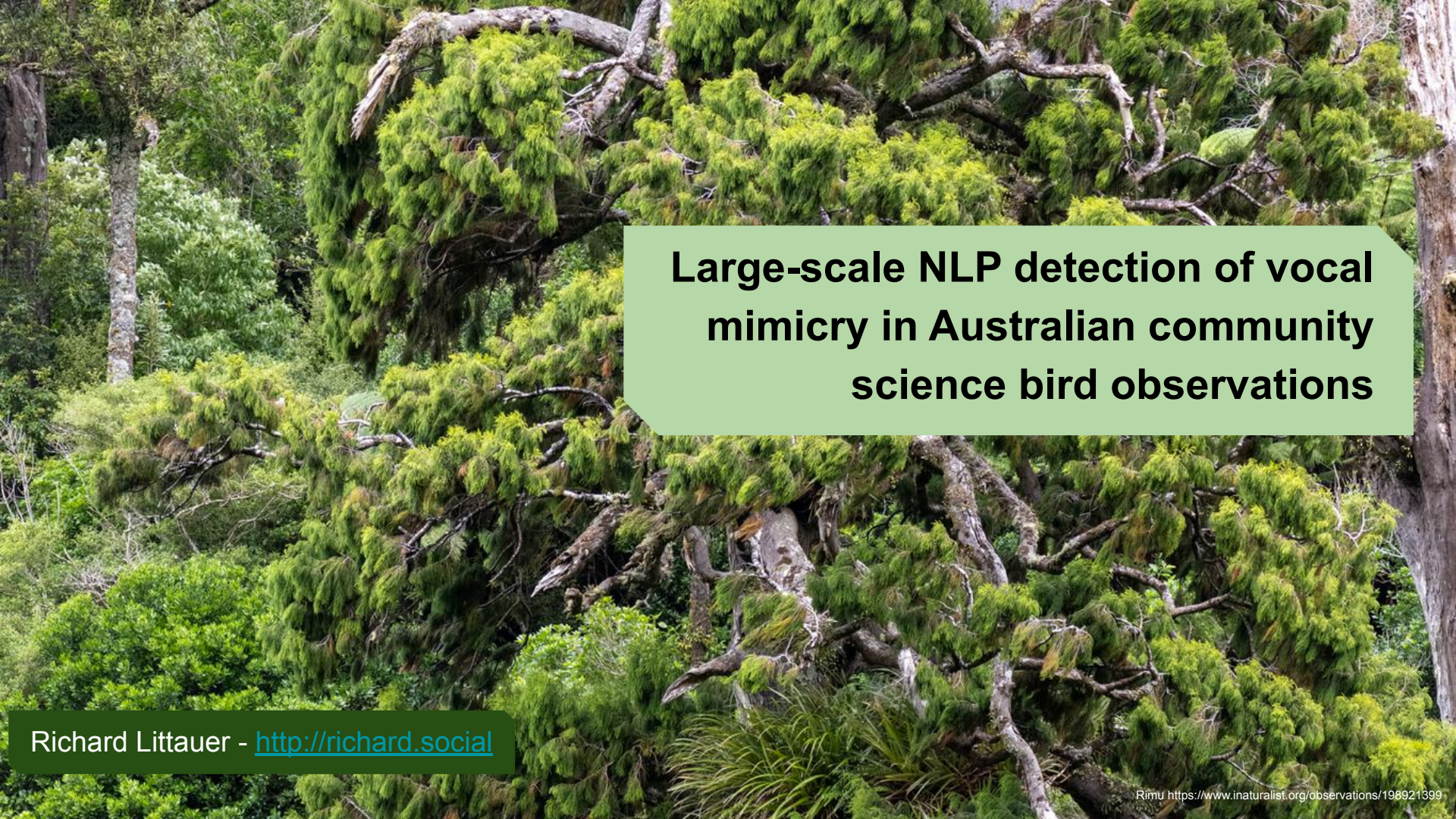
Large-scale NLP detection of vocal mimicry in Australian community science bird observations



Richard Littauer - http://richard.social

Rimu https://www.inaturalist.org/observations/198921399

Who is this Richard Littauer, anyway

- PhD Student
- Linguist
- Computer scientist
- Taxonomic nomenclaturist

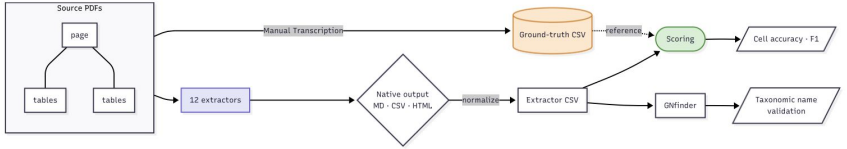


Fig. 3. The main PDF extractor pipeline.





eBird Stats WORLD

10,984
Species

131,804,198
Complete Checklists

1,370,246
eBirders

Top Regions WORLD | CURRENT MONTH

Checklists Leaders

1.	United States	315,687
2.	Canada	62,412
3.	Australia	31,114
4.	India	23,903
5.	United Kingdom	21,790
6.	Spain	15,797
7.	Brazil	12,987
8.	Colombia	7,894
9.	Taiwan	6,220
10.	Mexico	5,749

View All

Species Leaders

1.	Colombia	1,472
2.	Brazil	1,453
3.	Peru	1,413
4.	Ecuador	1,299
5.	Indonesia	1,204
6.	China	934
7.	India	889
8.	United States	800
9.	Mexico	791
10.	Bolivia	780

View All

Explore Species

Enter species name

Explore Regions

Enter a region

Observations

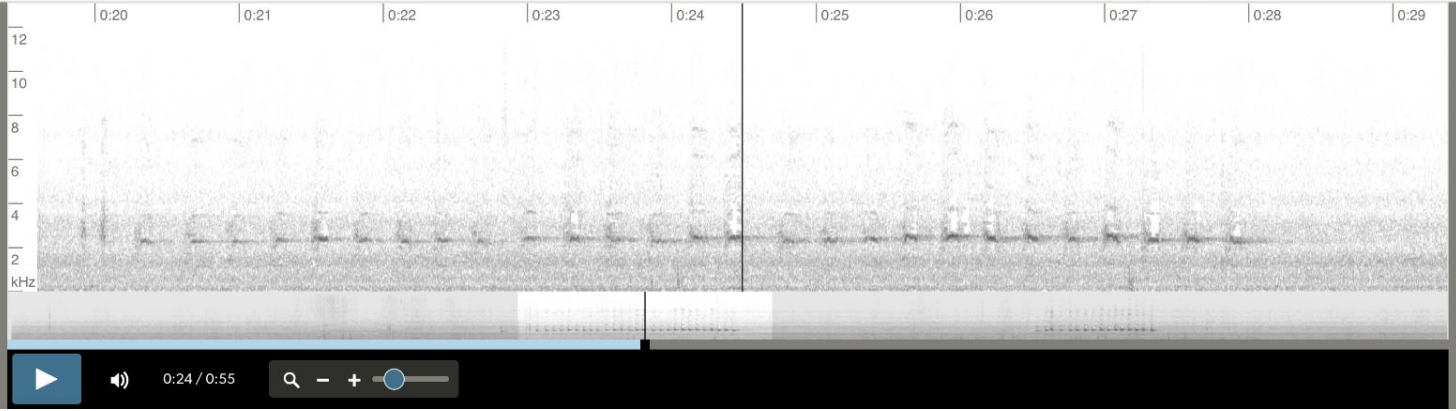
Species | Location | Go | Filters



About | Our Blog | DONATE | STORE

ANDROID APP ON Google Play


Canada Jay
Perisoreus canadensis



ML609943598

Canada Jay *Perisoreus canadensis*

★★★☆☆ 1 rating

+ Rate quality

Share Embed Report

Kim Selbee

Media from this contributor | Profile

14 Oct 2023

eBird checklist S152239877

Mayo

Yukon, Yukon Territory, Canada

Media from this location | Illustrated Checklist

Age Not specified

Sex Not specified

Playback Not specified

Observation details

2,1,1. One of the pair of jays at the river confluence was doing such a good imitation of a Northern Goshawk, that I was nearly fooled... I'm thinking they had very recently seen one!

Technical information

Recorder

Microphone

Accessories

Original file size 451.55 KB



Observation details

2,1,1. One of the pair of jays at the river confluence was doing such a good imitation of a Northern Goshawk, that I was nearly fooled... I'm thinking they had very recently seen one!

Mayo
Yukon, Yukon Territory, Canada
Media from this location | Illustrated Checklist

Observation details
2,1,1. One of the pair of jays at the river confluence was doing such a good imitation of a Northern Goshawk, that I was nearly fooled... I'm thinking they had very recently seen one!

Original file size 431.55 KB

Canada Jay *Perisoreus canadensis*

Dan Strickland and Henri R. Ouellet
Version: 1.0 — Published March 4, 2020
Text last updated November 8, 2018

Introduction
Field Identification
Plumages, Molts, and Structure
Systematics
Distribution
Habitat
Movements and Migration
Diet and Foraging
Sounds and Vocal Behavior
Vocalizations
Nonvocal Sounds
Behavior
Breeding
Demography and Populations

Only 8 species!

Sounds and Vocal Behavior

Vocalizations

Mimicry

The Canada Jay is previously reported to imitate Northern Pygmy-Owl (*Glaucidium gnoma*; 72), Rough-legged Hawk (*Buteo lagopus*) and Merlin (*Falco columbarius*; 29), and Pine Grosbeak and Red-tailed Hawk (*Buteo jamaicensis*; 93). Also gives excellent imitations of Broad-winged Hawk (*Buteo platypterus*), American Crow (*Corvus brachyrhynchos*), and Blue Jay, sometimes when the imitated species is in sight, but more often in plausibly stressful situations (e.g., human near active nest; bird just released after banding; DS). Imitation of Merlin's high-pitched tremolo was conspicuous in 2001–2002 on Anticosti Island, Quebec, where this predator is common. Merlin imitations were unknown before that in Algonquin Park, but increasingly heard since then, perhaps resulting from local recovery of formerly extirpated Merlin population (DS). As with Blue Jay (163), most species imitated by the Canada Jay are known or potential predators of adults or nests; mimicking predators could serve as a warning call to other group members, to confuse the predator itself, or merely to signal in a general way that a threat is present.

ounds

local nonbreeders sometimes accompanied by bill snapping (157; T.
unication).

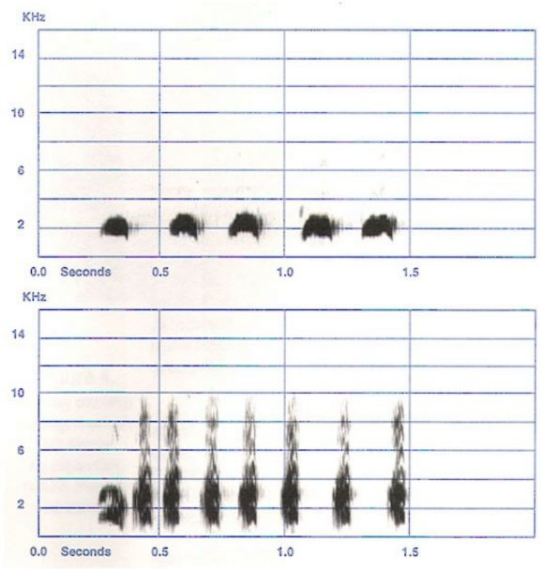


Figure 4. Canada Jay vocalizations. + Enlarge

A sample of Canada Jay vocalizations: (top) variant of Single Notes, often used by birds in groups to maintain contact; (bottom), subdued Chatter, often associated with the mobbing of predators. Newfoundland. Borror Lab of Bioaccoustics recording BLB–G93.



Avian vocal mimicry

“cases of vocal resemblance where a receiver specifically selects for resemblance” (Dalziell et al. 2015) between avian vocalizations of the mimic and the model species (Kelley et al. 2008)

And…

where that resemblance confers a benefit on the producer.



Vocal mimicry using comments?

- All 3 *Perisoreus* jays
- 450k observations from: eBird, iNaturalist, Xeno-Canto
- Only use comments: “imitate,” “mimic,”, “call”, banding codes

- Total retained: 234 observations



Vocal mimicry using comments?

- 40 species for *P. canadensis*
  - (+28 new species!)
- 20 for *P. internigrans*
- 0 for *P. internigrans*



Owenstrickland © CC-By https://www.inaturalist.org/observations/148011607



Riyga25 © CC-BY https://www.inaturalist.org/observations/321817995

The Wilson Journal of Ornithology ›
Latest Articles

Open access

751 Views | 0 CrossRef citations to date | 0 Altmetric

Listen

Short Communication

Community-science sourced evidence of vocal mimicry in *Perisoreus* jays

Preuves issues de la science participative attestant de l'imitation vocale chez les mésangeais (genre *Perisoreus*)

Richard Littauer, Sam Darmstadt, William M. Kirsch & Davyd H. Betchkal

Received 08 Nov 2024, Accepted 09 Jan 2026, Published online: 12 Feb 2026

Cite this article

https://doi.org/10.1080/15594491.2026.2616578

Check for updates

Full Article — Showing

Figures & data

References

Supplemental

Citations

Metrics

Licensing

Reprints & Permissions

PDF EPUB

Article contents | Related research

Australia?

- 70,000,000~ observations.
- Not an expert at recognizing birds here.
- This could be parachute science.



Filtering

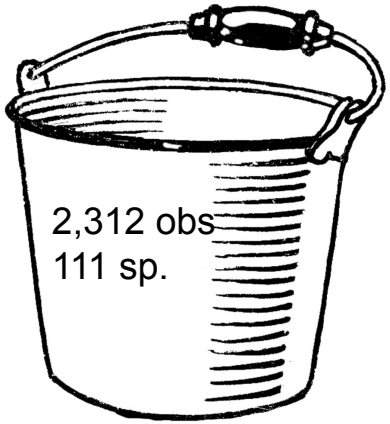
False positives: "resembled American Robin", "similar call to Woody the Woodpecker", "calling like crazy", "mock attacking"



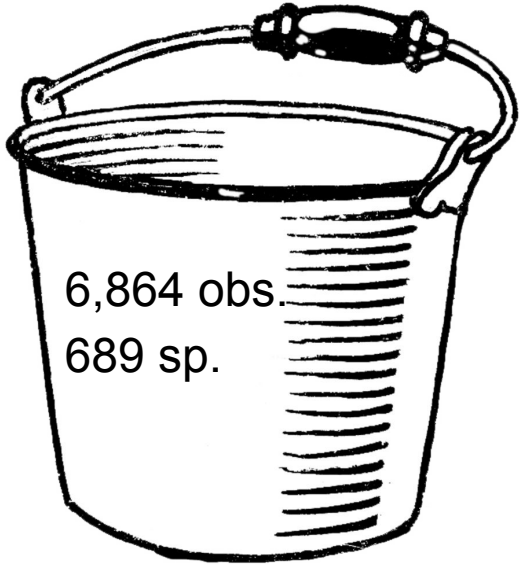


… and some Google Sheets work.

Results!



Validated.



Not validated.

Results!



Bihnz © CC-BY
https://www.inaturalist.org/observations/177752681

TOP 20 SPECIES BY NUMBER OF MIMICRY OBSERVATIONS:

Superb Lyrebird	596
European Starling	230
Olive-backed Oriole	216
Singing Bushlark	162
Spangled Drongo	106
Chestnut-rumped Heathwren	105
Brown Thornbill	103
Australian Magpie	103
Silvereye	99
Satin Bowerbird	79
Gray Shrikethrush	57
Sahul Cicadabird	56
Inland Thornbill	55
Albert's Lyrebird	52
Eastern Whipbird	51
Australasian Figbird	51
Australian Boobook	50
Gray Butcherbird	48
Yellow-tailed Black-Cockatoo	47
Pacific Koel	47



Saryu Mae 前 朝琉 Invertebratist © CC-BY
https://www.inaturalist.org/observations/261221876

Species with core mimicry mentions (111 species):

Species	Observations
Superb Lyrebird	579
European Starling	220
Olive-backed Oriole	200
Singing Bushlark	142
Spangled Drongo	92
Chestnut-rumped Heathwren	88
Silvereye	69
Brown Thornbill	69
Satin Bowerbird	66
Inland Thornbill	49
Albert's Lyrebird	49
Yellow-throated Scrubwren	34
Redthroat	32
Gray Shrikethrush	31
Brown Quail	30
Gray Butcherbird	27
Eurasian Blackbird	27
Regent Honeyeater	23
Australian Magpie	22
Lewin's Rail	22

Isabelline Wheatear, New Holland Honeyeater, Western Yellow Robin, Weebill, Bell Miner, Banded Whiteface, Latham's Snipe, Fan-tailed Cuckoo, Noisy Miner, White-eared Honeyeater, Pale-headed Rosella, Tawny Grassbird, Golden Whistler, Crescent Honeyeater, Black-faced Cuckooshrike, Sulphur-crested Cockatoo, Restless Flycatcher, Helmeted Friarbird, Brown Gerygone, Lewin's Honeyeater, Comb-crested Jacana, Spotless Crake, Fairy Gerygone, Spotted Quail-thrush, Varied Sittella, lark sp., Spinifexbird, Scarlet Myzomela, Alexandrine Parakeet, Rose-ringed Parakeet, Stubble Quail, Metallic Starling, Western Shrike-tit, Cockatiel, Large-billed Scrubwren, Chestnut-rumped Thornbill, White-eared Monarch, Rainbow Bee-eater, Northern Shrike-tit, Spiny-cheeked Honeyeater, Noisy Scrub-bird, Spotted Scrubwren, Blue-faced Honeyeater, Western Fieldwren



Mikhail_golomysov © CC-BY
https://www.inaturalist.org/observations/286727692

Australian birds mimic!

And questions remain about how much, and for what purpose.

But: community science comments can be used to source mimicry events.

Questions?

